Tutorial 3

Ex. 8.5

Q40 $\displaystyle\int \frac{e^{4t}+2e^{2t}-e^{t}}{e^{2t}+1}\,dt$

$u = e^t$

$du = e^t\,dt$

$$= \int \frac{\boxed{e^t}(e^{3t}+2e^{t}-1)}{e^{2t}+1}\,\boxed{dt}$$

$$= \int \frac{u^3+2u-1}{u^2+1}\,du$$

$$u^3+2u-1 = \boxed{u^3+u}+u-1 \Rightarrow \frac{u^3+2u-1}{u^2+1}$$

$$u(u^2+1) = u^3+u$$

$$= u + \frac{u-1}{u^2+1}$$

$$= u + \frac{u}{u^2+1} - \frac{1}{u^2+1}$$

Q46 $\displaystyle\int \frac{1}{(x^{1/3}-1)\sqrt{x}}\,dx$

Hint: ~~$x = u^6$~~

$u = \sqrt{x} \Rightarrow du = \dfrac{1}{2\sqrt{x}}\,dx$

$\Downarrow$

$x = u^2$

$x^{1/3} = u^{2/3}$

$t = u^{1/3} \Rightarrow u^{2/3} = t^2$

$dt = \frac{1}{3}u^{-2/3}\,du$

$= \frac{1}{3}(u^{1/3})^{-2}\,du$

$= \frac{1}{3}t^{-2}\,du$

~~dt~~ $du = \boxed{3t^2\,dt}$

$$= 2\int \frac{1}{u^{2/3}-1}\,du$$

$$= 2\int \frac{3t^2}{t^2-1}\,dt$$

$$= 6\int \frac{t^2}{t^2-1}\,dt$$

$$\frac{t^2}{t^2-1} = \frac{\boxed{t^2-1}+1}{t^2-1}$$

$$= 1 + \underbrace{\frac{1}{t^2-1}}$$

$$= \frac{1}{(t-1)(t+1)} = \frac{A}{t-1} + \frac{B}{t+1}$$

If $x^{m/n}$ is in integral, try $u = \boxed{x^{1/n}}$

$$du = \frac{1}{n} x^{-\frac{(n-1)}{n}} dx$$

$$= \frac{1}{n} \left(x^{1/n}\right)^{-(n-1)} dx$$

$$= \frac{1}{n} u^{-(n-1)} dx$$

$$\int \frac{1}{(x^{1/3}-1)\, x^{1/4}} dx \qquad u = x^{1/4} \Rightarrow dx = 4u^3 du$$

$$= \int \frac{4u^3}{(u^{4/3}-1)\, u} du = \int \frac{4u^2}{u^{4/3}-1} du \qquad t = u^{1/3}$$

$$= 12 \int \frac{t^8}{t^4-1} dt$$

Q50 $\displaystyle\int \frac{1}{x^6(x^5+4)}\,dx \cdot \frac{x^4}{x^4}$

$$= \int \frac{x^4}{\underbrace{x^{10}}_{u^2}(\underbrace{x^5}_{u}+4)}\,dx$$

$u = x^5$

$du = 5x^4\,dx$

$$= \frac{1}{5}\int \frac{1}{u^2(u+4)}\,dx$$

$$\frac{1}{u^2(u+4)} = \frac{A}{u} + \frac{B}{u^2} + \frac{C}{u+4}$$

Q49 $\displaystyle\int \frac{1}{x(x^4+1)}\,dx$

Hint: Mult. by $\dfrac{x^3}{x^3}$

$$= \int \frac{x^3}{\underbrace{x^4}_{u}(\underbrace{x^4}_{u}+1)}\,dx$$

Sub. $u = x^4$

$du = 4\boxed{x^3\,dx}$

$$\frac{1}{4}\cdot\frac{1}{u(u+1)}$$

$$\int \frac{1}{x^m(x^n+a)}\,dx$$

If $m+n-1$ is a multiple of $n$, $(m+n-1 = \ell n)$

try mult. by $\frac{x^{n-1}}{x^{n-1}}$.

$$\frac{1}{x^6(x^5+4)}$$

$6+5-1 = \underline{\underline{10}}$

Ex. 8.8

~~Q52~~ ① Just evaluate

② Direct Comparison Test

③ Limit Comparison Test

$\boxed{\int_a^\infty} f(x)\,dx$

$\rightarrow t - \sin t \geq 0$

Q46 $\int_0^1 \frac{dt}{t - \sin t}$

Hint: $t \geq \sin t$ for $t \geq 0$

$0 \leq \sin t$ for $0 \leq t \leq 1$.

$-\sin t \leq 0$

$t - \sin t \leq t$

$\frac{1}{t} \leq \frac{1}{t - \sin t}$ for ~~$t \geq 0$~~ $0 \leq t \leq 1$.

$\int_0^1 \frac{1}{t}\,dt \leq \int_0^1 \frac{1}{t - \sin t}\,dt$

$$\int_0^1 \frac{1}{t}\,dt = \lim_{c\to 0^+} \int_c^1 \frac{1}{t}\,dt$$

$$= \lim_{c\to 0^+} \left[\ln t\right]_c^1$$

$$= \lim_{c\to 0^+} \left[\underbrace{\ln 1}_{0} - \ln c\right] = \infty$$

When $c \to 0^+$, $\ln c \to -\infty$

$-\ln c \to +\infty$

Since $\int_0^1 \frac{1}{t}\,dt$ diverges to $\infty$ and $\int_0^1 \frac{1}{t}\,dt \leq \int_0^1 \frac{1}{t-\sin t}\,dt$,

$\int_0^1 \frac{1}{t-\sin t}\,dt$ diverges.

Q52 $\int_4^\infty \frac{dx}{\sqrt{x}-1}$

Vibe: $\frac{1}{\sqrt{x}-1} \sim \boxed{\frac{1}{\sqrt{x}}}$

① Just evaluate: $u = \sqrt{x} - 1$ (to get a rational function)

# Tedious

② Direct Comp. Test: $\sqrt{x} - 1 \leq \sqrt{x}$ for $x \geq 4$

$$\Downarrow$$

$$\boxed{\boxed{\frac{1}{\sqrt{x}}} \leq \frac{1}{\sqrt{x}-1}} \quad \text{for } x \geq 4.$$

$\int_4^\infty \frac{1}{\sqrt{x}}\, dx = \cdots$ (Diverges)

Since $\int_4^\infty \frac{1}{\sqrt{x}}\, dx$ diverges, $\int_4^\infty \frac{dx}{\sqrt{x}-1}$ diverges by Direct Comparison Test.

③ Limit Comp. Test:

Let $f(x) = \frac{1}{\sqrt{x}-1}$, $g(x) = \frac{1}{\sqrt{x}}$.

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{\sqrt{x}}{\sqrt{x}-1} = \lim_{x\to\infty} \frac{1}{1-\frac{1}{\sqrt{x}}} = \underline{1} > 0$$

$$\int_4^\infty \frac{1}{\sqrt{x}}\,dx = \cdots \quad (\text{Diverges})$$

Since $\int_4^\infty \frac{1}{\sqrt{x}}\,dx$ div., $\int_4^\infty \frac{1}{\sqrt{x}-1}\,dx$ div. by Limit Comparison Test.

Q53 $\frac{1}{\sqrt{v-1}} \sim \boxed{\frac{1}{\sqrt{v}}}$ div.

Q54 $\frac{1}{1+e^{\theta}} \sim \boxed{\frac{1}{e^{\theta}}}$ conv.

Q55 $\frac{1}{\sqrt{x^6+1}} \sim \frac{1}{\sqrt{x^3}} = \boxed{\frac{1}{x^{3/2}}}$ conv.

Q57 $\frac{\sqrt{x+1}}{x^2} \sim \frac{\sqrt{x}}{x^2} = \boxed{\frac{1}{x^{3/2}}}$ conv.

#Q59, Q60 won't work

Q65 $\frac{1}{\sqrt{e^x - x}} \sim \frac{1}{\sqrt{e^x}} = \boxed{\frac{1}{e^{x/2}}}$ conv.

Q66 $\frac{1}{e^x - 2^x} \sim \boxed{\frac{1}{e^x}}$ conv.